

Brazilian E-Commerce Analytics — dbt Assignment

Dataset

Use the Brazilian E-Commerce Public Dataset by Olist. The dataset represents a real e-commerce marketplace and includes customer, order, order-item, product, payment, seller, review, and product-category information. The files are related to each other and should be treated as a connected business dataset rather than as independent CSV files.

Dataset URLs

- Customers: https://raw.githubusercontent.com/Ganesh7699/Brazilian-E-Commerce-OList/master/olist_customers_dataset.csv
- Orders: https://raw.githubusercontent.com/Ganesh7699/Brazilian-E-Commerce-OList/master/olist_orders_dataset.csv
- Order Items: https://raw.githubusercontent.com/Ganesh7699/Brazilian-E-Commerce-OList/master/olist_order_items_dataset.csv
- Products: https://raw.githubusercontent.com/Ganesh7699/Brazilian-E-Commerce-OList/master/olist_products_dataset.csv
- Payments: https://raw.githubusercontent.com/Ganesh7699/Brazilian-E-Commerce-OList/master/olist_order_payments_dataset.csv
- Product Category Translation:
https://raw.githubusercontent.com/Ganesh7699/Brazilian-E-Commerce-OList/master/product_category_name_translation.csv

Assignment Scenario

You have joined the data engineering team of an e-commerce marketplace. The organization currently has raw customer, order, order-item, product, payment, and category information available in Snowflake, but the business team cannot directly use this raw operational data for reliable reporting. Your responsibility is to design and implement a dbt analytics project that converts the raw datasets into clean, trusted, and business-ready analytical data. The final solution should help management understand customer behaviour, sales trends, product and category performance, payment behaviour, regional performance, and order lifecycle information. While completing the project, you are expected to decide the appropriate project structure, model names, schemas, materializations, views, tables, and analytical outputs based on the business requirement.

Required Tasks

- Create a new **dbt project** for this e-commerce analytics solution and organize it using a meaningful project structure.
- Load or make the provided datasets available in a **raw Snowflake layer** before beginning the dbt transformations.

- Define all required raw datasets as **dbt sources** so that the transformations reference source definitions instead of directly hard-coding physical table locations.
- Create suitable **staging models** for the main datasets. The staging layer should clean and standardize the source information, rename unclear fields where necessary, handle missing values sensibly, and convert columns to suitable datatypes.
- Create at least one **intermediate transformation layer** that combines related datasets and prepares reusable business-level information for downstream analytical models.
- Use **ref()** **relationships** between dbt resources so that model dependencies are managed through dbt rather than by manually referring to physical tables.
- Create at least **three final business models or marts** that provide useful analytics for management. The marts should cover different perspectives such as customer performance, product/category performance, sales performance, geographical performance, payment behaviour, or order lifecycle analysis.
- Ensure that at least one final analytical model calculates a meaningful **business measure**, such as revenue, order value, number of orders, customer contribution, product performance, category performance, or another useful KPI derived from the available data.
- Decide which dbt models should be implemented as **views** and **which should be implemented as tables**. Be prepared to explain why you selected each materialization.
- Create at least one **dbt seed** containing a small piece of business-maintained reference data. The seed may represent sales targets, regional classifications, category targets, priority categories, or another useful business reference.
- Use the seed inside at least one dbt transformation so that the final analytical model combines **actual business data with the reference data from the seed**.
- Create appropriate **basic dbt tests** using only:
 - not_null
 - unique
 - accepted_values
 - relationships
- Apply the basic tests only where they make business sense. Decide which identifiers must be unique, which columns must not be null, which values should be controlled, and which parent-child relationships must remain valid.

- Create at least **two relationship tests** between logically related datasets. The relationship should represent real business relationships present in the e-commerce data.
- Introduce one temporary **data-quality problem** into the raw data, such as a missing identifier, duplicate identifier, invalid reference, or invalid controlled value, and demonstrate that one of the dbt tests detects the issue.
- Correct the intentionally introduced data-quality issue and confirm that the affected test passes after the correction.
- Identify one source dataset containing information that could realistically **change over time** and use it to create a **dbt snapshot**.
- Select an appropriate snapshot strategy based on the available source information and explain why that strategy is suitable.
- Simulate at least one meaningful business change to the selected source record and demonstrate that the snapshot preserves both the old and new versions.
- Show the difference between the **current source record** and the **historical snapshot records** after the change.
- Add meaningful **descriptions and documentation** for the important sources, models, and business-facing outputs created in the project.
- Ensure the dbt project produces a clear **lineage** showing the flow from raw sources through staging and intermediate transformations to the final marts.
- Use only the final analytical models to answer business questions such as:
 - Which product categories contribute the highest sales?
 - Which geographical regions contribute the most business?
 - Which customers contribute the highest value?
 - How does order volume change over time?
 - Which payment methods are most frequently used?
 - Which business areas are meeting or missing their target?
 - What changes can be identified through the snapshot history?